

Classifying Rational and Irrational Numbers

Fraction forms of rational numbers, roots and pi, and placing an irrational number on the number line

The test you are using

- A number is **rational** exactly when it can be written as a fraction $\frac{\text{integer}}{\text{integer}}$. Terminating decimals, repeating decimals, and roots that come out exact are all rational.
 - A number is **irrational** when no such fraction exists. Its decimal runs forever *without* settling into a repeating block.
 - **So, unless a problem says otherwise:** if the number is rational, prove it by writing the fraction in lowest terms; if it is irrational, write the value rounded as the problem asks.
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1. Write 0.35 as a fraction of two integers in lowest terms.

Answer: _____

2. Evaluate $\sqrt{121}$, and write the value as a fraction of two integers.

Answer: _____

3. $\sqrt{54}$ cannot be written as a fraction, so locate it instead.

(a) $\sqrt{54}$ lies between the whole numbers _____ and _____.

(b) To the nearest hundredth, $\sqrt{54} \approx$ _____.

(c) Mark and label your value with a dot on the number line below.



4. Show that the repeating decimal $0.\overline{6}$ is rational. Let $x = 0.\overline{6}$, multiply both sides by ten, subtract the original equation, solve the equation you are left with, and write x in lowest terms.

Answer: _____

5. Consider $\sqrt{169}$.

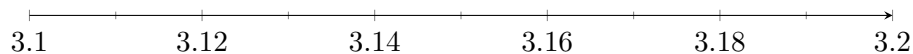
(a) Evaluate it. Answer: _____

(b) Explain, in one sentence, how the value you found shows that $\sqrt{169}$ is rational.

6. One of π and $\frac{22}{7}$ is irrational and one is rational, and they are very close together.

(a) Write $<$, $>$, or $=$ between them. π _____ $\frac{22}{7}$

(b) Mark and label both numbers with dots on the number line below.



7. Consider $\frac{9}{12}$.

(a) Write it in lowest terms. Answer: _____

(b) Write that same number as a decimal. Answer: _____

8. Evaluate $\sqrt{\frac{9}{4}}$ and write the value as a fraction of two integers in lowest terms.

Answer: _____

9. Write these four numbers in order from least to greatest.

$$\sqrt{7} \quad \sqrt{9} \quad \pi \quad \frac{10}{3}$$

Answer: _____

10. **Challenge.** Tomas says: “The digits of $0.1\overline{6}$ never stop, so it must be irrational.”

- (a) Write $0.1\overline{6}$ as a fraction in lowest terms. Answer: _____
- (b) Explain what is wrong with the reason Tomas gave.